# Naveen Saragadam

Senior Data Engineer | Azure | Databricks | AWS | Snowflake | Spark/PySpark | Python | SQL
**Email:** naveens0519@gmail.com | **Phone:** +1 (480) 869-6966 | **LinkedIn:** https://www.linkedin.com/in/naveen-saragadam

## Summary

- Senior Data Engineer with 10+ years of experience designing, developing, modernizing, and supporting enterprise data platforms, cloud data pipelines, ETL/ELT solutions, distributed processing frameworks, data lakes, data warehouses, and analytical environments across Azure, AWS, Databricks, Spark/PySpark, Python, SQL, and Hadoop.
- Extensive experience delivering end-to-end data engineering solutions spanning source-system integration, data ingestion, batch processing, streaming, transformation, data quality, data migration, data warehousing, analytics, production deployment, testing, troubleshooting, and operational support.
- Strong Azure engineering background with hands-on experience across Azure Data Factory, Azure Data Lake Storage, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, Azure HDInsight, Azure Blob Storage, Azure Key Vault, Azure DevOps, Azure Cosmos DB, and Azure Analysis Services.
- Strong AWS experience across Amazon S3, EC2, EMR, Redshift, Glue, Lambda, Step Functions, CloudWatch, Kinesis, SNS, SQS, DynamoDB, and Athena, supporting enterprise data ingestion, distributed processing, event-driven integration, and hybrid-cloud data workflows.
- Extensive Databricks and Apache Spark experience developing PySpark, Spark SQL, Scala, Spark Streaming, Spark DataFrame, and distributed processing applications for large-scale extraction, transformation, cleansing, aggregation, migration, and analytics workloads.
- Strong Hadoop ecosystem background including HDFS, Hive, MapReduce, Sqoop, Flume, Pig, Oozie, ZooKeeper, YARN, HBase, Cassandra, Impala, Hortonworks, and Cloudera, with experience modernizing legacy Hadoop and MapReduce workloads to Spark and cloud platforms.
- Experienced in developing and orchestrating enterprise ETL/ELT pipelines using Azure Data Factory, Informatica PowerCenter, Informatica IICS, SSIS, Talend, Airflow, Oozie, Python, SQL, shell scripting, and REST/HTTP integrations.
- Experienced in real-time and event-driven data processing using Apache Kafka, Amazon Kinesis, SNS/SQS, Google Cloud Pub/Sub, AWS DMS, Oracle GoldenGate, and Spark Streaming, including change data capture, messaging, incremental ingestion, and downstream processing.
- Strong background in enterprise cloud migration and modernization, including migration of on-premises SQL Server, relational database, Hadoop, and ETL workloads into Azure Data Lake, Azure SQL, Azure Synapse Analytics, Azure Databricks, Blob Storage, and Snowflake-based delivery environments.
- Strong SQL and database engineering experience across SQL Server, Oracle, MySQL, PostgreSQL, Teradata, DB2, Sybase, Snowflake, HBase, Cassandra, MongoDB, DynamoDB, and Azure Cosmos DB, including complex queries, stored procedures, views, triggers, indexing, and performance optimization.
- Proficient in Python, SQL, PySpark, Scala, Java, Shell scripting, HiveQL, PL/SQL, automation scripting, JSON deployment scripts, and UDF development for enterprise data engineering and operational automation.

## Skills

**Programming::** Python, SQL, PySpark, Scala, Java, Shell scripting, Bash, HiveQL, PL/SQL, NumPy
**Data Engineering / ETL::** ETL/ELT, data ingestion, data integration, data migration, batch processing, incremental processing, CDC, data transformation, source-to-target integration, Azure Data Factory, Informatica PowerCenter, Informatica IICS, SSIS, Talend, Airflow, Oozie, Sqoop, Flume
**Azure Data Services::** Azure Data Factory, Azure Databricks, Azure Data Lake Storage (ADLS/ADLS Gen2), Azure Synapse Analytics, Azure SQL Database, Azure SQL Data Warehouse, Azure Blob Storage, Azure HDInsight, Azure Key Vault, Azure DevOps, Azure Cosmos DB, Azure Analysis Services, Medallion architecture (Bronze/Silver/Gold)
**AWS Data Services::** Amazon S3, Amazon EC2, Amazon EMR, Amazon Redshift, AWS Glue, AWS Lambda, AWS Step Functions, Amazon CloudWatch, Amazon Kinesis Data Streams, Kinesis Data Firehose, Kinesis Video Streams, Amazon SNS, Amazon SQS, DynamoDB, Athena
**Databricks / Apache Spark::** Azure Databricks, Apache Spark, Spark Core, Spark SQL, PySpark, Spark Streaming, Spark DataFrames, RDDs, Scala, Spark UDFs, distributed processing, caching, partitioning, parallelism, memory optimization, workload tuning, Auto Loader
**Hadoop Ecosystem::** Hadoop, HDFS, Hive, HiveQL, MapReduce, Sqoop, Flume, Pig, Oozie, ZooKeeper, YARN, HBase, Cassandra, Impala, Hortonworks, Cloudera, Cloudera Manager
**Streaming / Messaging / CDC::** Apache Kafka, Kafka topics, producers, consumers, brokers and partitions, Spark Streaming, Amazon Kinesis, SNS, SQS, Google Cloud Pub/Sub, AWS DMS, Oracle GoldenGate, event-driven data integration, change data capture
**Databases / Warehousing / Modeling::** Snowflake, SQL Server, Azure SQL Database, Oracle, MySQL, PostgreSQL, Teradata, DB2, Sybase, Redshift, HBase, Cassandra, MongoDB, DynamoDB, Azure Cosmos DB, data lakes, data warehouses, data marts, data hubs, master data, staging/core layers, facts and dimensions, relational data models, analytical datasets, enterprise reporting layers, Parquet, Avro, JSON, ORC, SequenceFile, CSV, structured data, semi-structured data
**Analytics / BI::** Power BI, Tableau, Spotfire, Cognos, OBIEE, SSAS, SSRS, SSDT, Azure Analysis Services

**DevOps / Deployment / Reliability::** Azure DevOps, Visual Studio Team Services (VSTS), Git, CI/CD, build and release processes, JSON deployment scripts, automation scripting, Linux, Unix, Windows, Agile, Scrum, SDLC, Test-Driven Development, production support, release management
**Data Quality / Reliability::** Data validation, business rules, data-quality analysis, functional testing, system integration testing, end-to-end testing, regression testing, acceptance testing, smoke testing, disaster recovery, geo-replication, regional failover, production acceptance, pre-production testing

## Certifications

- Microsoft Certified: Azure Databricks Data Engineer Associate
- Microsoft Certified: Fabric Data Engineer Associate
- AWS Certified Data Engineer – Associate
- Databricks Certified Data Engineer Associate

## Work Experience

### JPMorgan Chase
**Austin, TX | Feb 2024 – Present**

Senior Data Engineer | Azure
Hybrid Contract

- Design and implement enterprise data pipelines using Azure Data Factory, Azure Databricks, PySpark, Spark SQL, T-SQL, ADLS, Azure SQL, Azure Synapse Analytics, and Azure Blob Storage to support production data integration and analytical workloads.
- Develop Azure Data Factory pipelines using linked services, datasets, Mapping Data Flows, triggers, and pipeline activities to automate ingestion, transformation, validation, and delivery of enterprise datasets.
- Build scalable Databricks and PySpark ETL workflows for large datasets, applying partitioning, parallel processing, caching, and Spark optimization techniques to improve processing efficiency and reliability.
- Implement incremental and event-driven ingestion patterns using Databricks Auto Loader for files arriving in Azure Data Lake Storage.
- Support migration of on-premises SQL Server workloads to Azure-based data platforms including Azure SQL Database and Azure Synapse Analytics, redesigning ingestion and transformation workflows for cloud execution.
- Integrate Snowflake with Azure data workflows to deliver curated datasets for downstream analytical consumption.
- Implement database reliability and disaster-recovery capabilities using Azure SQL geo-replication, failover strategies, elastic pools, and elastic jobs where appropriate for production workloads.
- Develop SQL scripts, stored procedures, joins, subqueries, and data validation logic to support transformation, reconciliation, and analytical data processing.
- Build Python and SQL automation utilities for operational tasks, data validation, and pipeline support, reducing manual intervention across recurring data-engineering activities.
- Support streaming and CDC-based data integration using Apache Kafka, AWS DMS, Oracle GoldenGate, and related ingestion patterns where required by source and downstream systems.
- Develop and maintain data-quality checks and validation processes to identify missing, duplicate, invalid, and inconsistent records before downstream consumption.
- Implement CI/CD and deployment processes using Azure DevOps and JSON-based deployment configurations for repeatable release of data pipelines and platform changes.
- Troubleshoot production data pipelines, investigate failures and data discrepancies, and work with application, database, analytics, and business teams to resolve data issues and maintain delivery SLAs.
- Support more than 35 production pipelines and workflows processing approximately 2 TB of data per day across multiple enterprise source systems and downstream datasets.
- Improve pipeline processing time by approximately 20% through Spark tuning, partitioning, query optimization, and workflow improvements.
- Reduce recurring manual operational effort by approximately 30% through Python/SQL automation and improved pipeline validation.
- Load curated data from Azure Data Lake into Blob Storage for downstream Snowflake delivery and enterprise analytical consumption.
- Develop and consume RESTful web services to enable communication and data exchange between enterprise systems and data platforms.
- Develop Python solutions using core language features, data structures, algorithms, and reusable automation utilities to support data-processing requirements.
- Work with Apache Kafka topics, producers, consumers, brokers, and partitions in support of event-driven data integration and messaging requirements.
- Work with Amazon Kinesis Data Streams, Kinesis Data Firehose, and Kinesis Video Streams as part of cloud-based streaming and integration workflows.

- Work with change-data-capture technologies including Kafka, AWS DMS, and Oracle GoldenGate to support incremental data movement and synchronization.
- Perform data-quality analysis and apply business rules across extraction, transformation, and loading layers.
- Validate solutions through functional, system integration, end-to-end, regression, sanity, acceptance, smoke, disaster-recovery, production-acceptance, and pre-production testing.
- Develop ETL jobs using REST/HTTP APIs, Java, and Talend to load data into HDFS and connect cloud-based workflows with Hadoop environments.
- Develop UDFs in Scala and PySpark to address specialized transformation and business requirements.

**Environment:** Azure Data Factory, Azure Databricks, Azure Synapse Analytics, Azure SQL Database, Azure Data Lake Storage, Azure Blob Storage, Snowflake, Azure Analysis Services, Azure Key Vault, SQL Server, Kafka, Amazon Kinesis, AWS SNS/SQS, Google Cloud Pub/Sub, AWS DMS, Oracle GoldenGate, REST APIs, Python, PySpark, Spark SQL, Scala, Java, Talend, T-SQL, SSDT, VSTS/Azure DevOps.

### GEICO
**Maryland | Oct 2021 – Jan 2024**

Data Engineer

- Develop scalable data pipelines using Azure Data Factory, Azure Databricks, Spark SQL, PySpark, T-SQL, and Azure Data Lake for enterprise data integration and analytics.
- Design data-lake and warehouse workflows supporting centralized data storage, master data, security, data hubs, data marts, and downstream analytical workloads.
- Develop distributed data transformations using Hadoop, Hive, MapReduce, Spark, PySpark, and Shell scripting to process large volumes of structured and semi-structured data.
- Modernize legacy MapReduce transformations by implementing equivalent Spark and PySpark processing patterns to improve maintainability and execution efficiency.
- Develop Spark applications using Python, Scala, and Spark SQL for batch transformation, aggregation, and analytical workloads.
- Implement Spark Streaming pipelines for ingesting and processing continuously arriving data for downstream Hive and analytical workloads.
- Create Hive external tables and optimize partitioning, bucketing, joins, and data layouts to improve query and processing performance.
- Develop Talend-based ETL workflows for integrating data from enterprise source systems into centralized data platforms.
- Optimize warehouse queries, data-processing jobs, and storage strategies to improve performance for production workloads.
- Develop SQL-based transformations and data validation logic using T-SQL, Spark SQL, and related data-engineering technologies.
- Support Informatica-based ETL processes and collaborate with development and testing teams during production releases and validation activities.
- Develop and maintain Power BI data sources and curated datasets to support enterprise reporting and analytics.
- Troubleshoot production failures, investigate data-quality issues, and support root-cause analysis across data pipelines and downstream applications.
- Support more than 25 production pipelines processing approximately 1.5 TB of data per day across enterprise datasets.
- Improved processing performance by approximately 25% through Spark optimization, partitioning, SQL tuning, and transformation redesign.
- Extracted, transformed, and loaded source data into Azure using Azure Data Factory, T-SQL, Spark SQL, and Azure Data Lake Analytics.
- Developed Hive queries against heterogeneous data-lake sources and persisted processed datasets to HDFS for downstream SQL access.
- Loaded data into Azure through external-table approaches and developed Azure SQL Data Warehouse solutions for forecasting and budgeting.
- Documented target-system impacts across data landing, staging, core, data-sharing, and visualization layers.
- Built Azure data-warehouse datasets supporting Power BI reporting and analytical consumption.
- Led estimation and participated in technical reviews, identifying implementation complexities and communicating project impacts to stakeholders.
- Supported cloud migration and modernization activities involving Hadoop, legacy ETL, SQL workloads, and Azure data platforms.

**Environment:** Azure Data Factory, Azure Databricks, Azure Data Lake, Azure Data Lake Analytics, Azure SQL Data Warehouse, Azure Stream Analytics, Hadoop, HDFS, Hive, MapReduce, Spark, PySpark, Spark SQL, Spark Streaming, Scala, Python, Talend, Informatica, shell scripting, SQL, Power BI, OBIEE, Agile/Scrum.

### AT&T
**Dallas, TX | May 2018 – Sep 2021**

Big Data Engineer

- Develop Big Data solutions integrating enterprise relational sources with Hadoop and cloud-based data platforms using Sqoop, Hive, Spark, PySpark, Azure Data Factory, and Azure storage services.

- Build data-ingestion workflows to move DB2 and Oracle data into HDFS and Hive using Sqoop and automated batch-processing jobs.
- Develop Flume, Sqoop, Pig, and MapReduce workflows for collecting, transforming, and loading large datasets into HDFS-based data lakes.
- Develop and maintain Hive external tables, dynamic partitions, and bucketing strategies to support efficient storage and analytical processing.
- Modernize MapReduce workloads by developing Spark and Scala transformations for distributed data processing.
- Develop Spark SQL and PySpark ETL pipelines for data transformation, cleansing, aggregation, and downstream analytical consumption.
- Support Hadoop cluster coordination and workflow execution using ZooKeeper, YARN, Hive, HBase, and related platform technologies.
- Develop Azure Data Factory pipelines and JSON deployment configurations to automate data movement between web APIs, Blob Storage, Azure SQL, and Azure analytical platforms.
- Implement data integration workflows moving data from Blob Storage and ADLS into Azure SQL and Azure SQL Data Warehouse / Azure Synapse Analytics.
- Develop REST-based data ingestion workflows and integrate web APIs with downstream cloud data stores.
- Develop ETL processes using SSIS, Azure Data Factory, Shell scripting, PowerShell, and SQL to support enterprise data movement and transformation.
- Work with Apache Kafka and Apache NiFi to support distributed data ingestion and streaming-oriented data flows.
- Implement data processing solutions using Cloudera Hadoop technologies including HDFS, Hive, Pig, HBase, YARN, and related ecosystem components.
- Support more than 25 source systems and approximately 1 TB of data per day as part of enterprise Big Data and cloud modernization initiatives.
- Reduced processing time by approximately 30% through migration from legacy MapReduce workloads to Spark-based distributed processing.
- Designed daily incremental ingestion processes to move raw DB2 data into Hive using Sqoop.
- Debugged and optimized MapReduce jobs using MRUnit and distributed-processing performance techniques.
- Used Oozie and ZooKeeper to schedule, coordinate, and monitor Hadoop workflows.
- Performed data migration using SQL, Azure SQL, Azure Storage, Azure Data Factory, SSIS, and PowerShell.
- Helped build a Hadoop data lake on Cloudera and developed campaign preprocessing and opportunity-generation pipelines using Hive and Spark.
- Developed Scala proofs of concept on YARN and compared Spark performance with Hive and SQL/Teradata workloads.
- Collaborated with infrastructure teams on Kafka- and Storm-based data-processing solutions.
- Worked with Hadoop ecosystem components including Hive, Pig, HBase, HBase-Hive integration, PySpark, Sqoop, and Flume.
- Developed and maintained SQL, Python, Scala, shell, and deployment automation supporting enterprise data workflows.
- Supported testing, troubleshooting, production deployment, and performance optimization of distributed data applications.

**Environment:** Hadoop, HDFS, Hive, Impala, Oracle, DB2, Spark, Spark SQL, PySpark, Scala, Sqoop, Oozie, Flume, Pig, MapReduce, Teradata, Kafka, ZooKeeper, HBase, NiFi, Solr, webMethods, Cloudera, YARN, Azure Data Factory, Azure SQL, Azure Storage, SSIS, PowerShell, REST APIs.

### UnitedHealth Group

**San Francisco, CA | Jan 2016 – Apr 2018**

ETL Developer

- Develop Hadoop-based ETL pipelines using Sqoop, Hive, Spark, PySpark, Python, Shell scripting, and Oozie for large-scale data processing and integration.
- Build ingestion workflows to move relational data from Oracle and SQL Server into Hadoop using Sqoop and automated batch-processing jobs.
- Develop Hive-based transformation and aggregation workflows for structured and semi-structured enterprise datasets.
- Develop Spark applications using PySpark and Spark SQL for distributed data transformation and processing.
- Process data using Hadoop YARN and Spark-on-YARN environments and support distributed batch execution across enterprise workloads.
- Use Avro, Parquet, JSON, and related data formats for efficient storage and processing of large datasets.
- Develop and maintain Kafka-based data ingestion workflows supporting streaming-oriented data integration requirements.
- Build Oozie and Airflow workflows to orchestrate ETL jobs, dependencies, schedules, and recurring data-processing tasks.
- Develop data-quality validation, reconciliation, and exception-handling processes across ETL workflows.
- Support traditional ETL processes alongside Hadoop-based processing to integrate relational and distributed data platforms.
- Investigate production failures, resolve data discrepancies, and collaborate with application and database teams to maintain reliable data delivery.

- Support more than 20 recurring ETL workflows processing approximately 500 GB of data per day across multiple source systems.
- Improved processing efficiency by approximately 20% through Spark transformation optimization, data-layout improvements, and workflow tuning.
- Processed Avro, Parquet, and semi-structured JSON data and converted JSON datasets to Parquet using PySpark DataFrames.
- Analyzed system failures, identified root causes, recommended corrective actions, and documented procedures for future operational use.
- Developed workflows using Sqoop, HiveQL, Spark, and Kafka to ingest enterprise messaging and relational data into HDFS.
- Developed Oozie and Airflow workflows to automate HDFS loading, preprocessing, scheduling, and recurring data-processing operations.
- Integrated Hadoop with traditional ETL solutions to process large volumes of semi-structured and unstructured data in HDFS.
- Supported performance tuning, production troubleshooting, deployment, data validation, and operational reliability of enterprise data pipelines.

**Environment:** Hadoop, HDFS, Hive, HiveQL, Sqoop, Spark, PySpark, YARN, MRv2, Kafka, Oozie, Airflow, Pig, HBase, Impala, Hortonworks, Cloudera Manager, Amazon EMR, Python, Java, Shell scripting, Avro, Parquet, JSON, Oracle, SQL Server, ETL.

### NCS Pearson India Private Limited
**Delhi, India | Mar 2013 – Apr 2014**

Hadoop Developer Intern

- Supported development and administration of distributed Hadoop environments for Big Data proof-of-concept and data-processing initiatives.
- Developed Hive-based data-processing workflows for structured data stored in HDFS.
- Developed MapReduce programs for distributed data transformation and batch processing.
- Used Sqoop and Flume to ingest data from relational and external sources into HDFS.
- Developed Pig scripts and UDFs for data transformation and preprocessing.
- Worked with HDFS, Hive, HBase, Sqoop, Flume, Pig, ZooKeeper, Oozie, and related Hadoop ecosystem technologies.
- Developed scripts using Python, Java, PL/SQL, and UNIX Shell to automate recurring data-processing and operational tasks.
- Installed and configured fully distributed Hadoop clusters across multi-node environments.
- Built data-ingestion pipelines using HDInsight, Flume, Sqoop, and Pig to extract weblog data and store it in HDFS.
- Developed Hive queries for data sampling, transformation, and analytical processing.
- Identified PL/SQL batch applications in General Ledger processing and compared performance to assess potential Hadoop migration benefits.
- Implemented proofs of concept demonstrating the potential benefits of migrating selected workloads to Hadoop.

**Environment:** Red Hat Linux, SQL Server, Oracle, MongoDB, Hadoop, HDFS, Hive, MapReduce, Pig, HBase, Sqoop, Flume, ZooKeeper, Oozie, Tez, Python, Java, PL/SQL, UNIX shell scripting, Cloudera Manager, HDInsight.

## Education

### University of Arizona
**Tucson, AZ | Dec 2015**

Master of Science in Computer Science

### KMITE College
**India | May 2014**

Bachelor of Technology in Computer Science